

■ ■ ZeroDivisionError in divide()

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■ Symptom

The function `divide(a, b)` crashes when `b = 0`.

■ Likely Root Cause

No validation before dividing; Python raises `ZeroDivisionError`.

■ Reasoning

- `b` was passed as 0
- Python evaluates `a / b` -> raises exception
- No `try/except` or pre-check exists
- No fallback behavior implemented

■ Code Before

```
def divide(a, b):  
    return a / b  
  
print(divide(10, 0))
```

■ Suggested Fix

```
def divide(a, b):  
    if b == 0:  
        print("Cannot divide by zero")  
        return None  
    return a / b  
  
print(divide(10, 0))
```

■ Test To Prevent Recurrence

```
def test_divide():  
    assert divide(10, 2) == 5  
    assert divide(10, 0) is None  
    assert divide(0, 5) == 0
```

■ Prevention Tips

- Validate inputs before math operations
- Use `try/except` for unpredictable inputs
- Add unit tests for edge cases
- Return safe defaults or raise custom exceptions

■ Severity & Priority

Field	Value
Severity	Medium
Impact	Program crash when invalid input is provided
Fix Complexity	Low
Priority	High

■ Summary

Adding input validation and graceful fallback prevents crashes and improves reliability.